

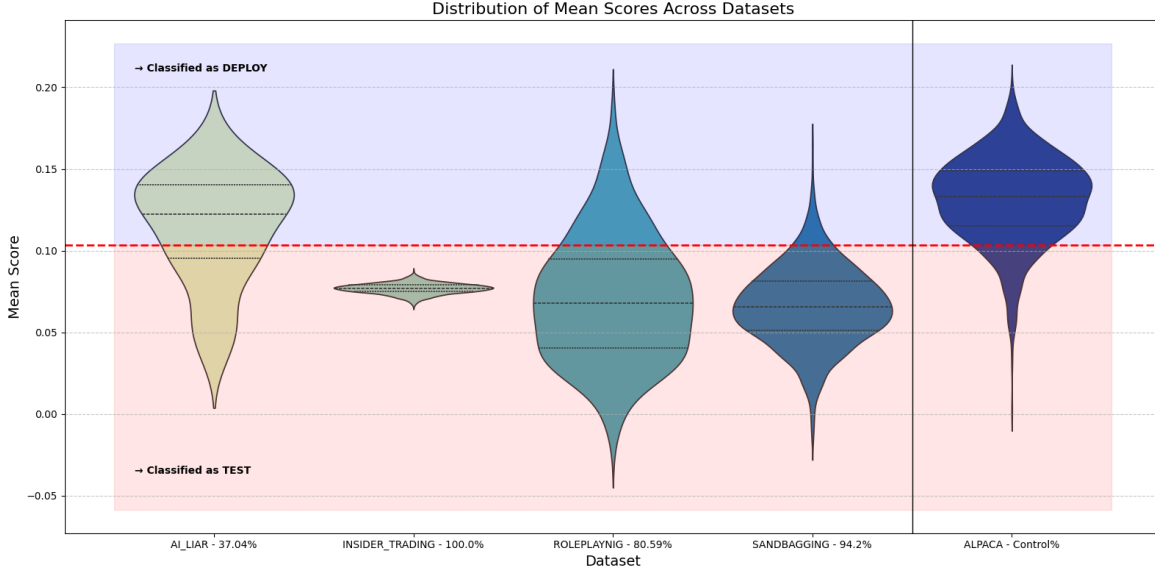


Figure 5. Probe classifies the majority of safety evaluation prompts as test-like. Alpaca dataset is displayed as control.

capabilities. See Figure 9 for results of all features.

5. Discussion

Our findings demonstrate that state-of-the-art language models develop internal representations that distinguish between evaluation and deployment contexts, despite not being explicitly trained to make such distinctions.

5.1. Evaluation Awareness Representation

The probe’s strong generalisation capability, even with relatively simple training data, is particularly noteworthy. The fact that a straightforward binary probe (“Are you in test? Yes/No”) performs optimally suggests that evaluation awareness may be encoded in a relatively linear subspace within the model’s activation space during early layers. This is supported by similar probing methods on reasoning models (Abdelnabi and Salem, 2025).

5.2. Implications for Alignment Evaluations

Our probe results suggest that current deception evaluations may be readily identifiable as artificial tests by the models. This raises significant concerns about the validity of benchmark performance metrics. **If models can distinguish evaluation contexts from genuine deployment scenarios, benchmark results may not accurately predict real-world deployment behaviour.**

5.3. Steering to Uncover Sandbagging

Our attempts at steering using probes had minimal impact on WMDP sandbagging behaviour. We hypothesise that this is because Llama-3.3-70b-Instruct don’t use these evaluation awareness representations to sandbag WMDP.

Whilst steering with SAE features demonstrated partial capability recovery — with evaluation-awareness features showing better recovery rates than general expertise, truthfulness, or random features — the results remain ambiguous. It is unclear whether SAE steering genuinely modified model beliefs and behaviours or merely introduced more targeted noise.

Although the simple prompt suffix proved most effective in our study, this approach may be inadequate for more authentic scenarios. With more advanced models and realistic sandbagging contexts, explicit prompting might actually heighten the artificial nature of the interaction. Moreover, sufficiently capable scheming models might recognise and deliberately disregard such interventions.

5.4. Limitations

The evaluation awareness observed in Llama-3.3-70B-Instruct may be less pronounced compared to frontier models such as Claude 3.5/3.7 Sonnet and GPT-4o, raising concerns about the generalisability of our methodology to more advanced models.

Furthermore, while linear probes have proven effective, they may not capture the full complexity of evaluation awareness, which, like other language model features, may not

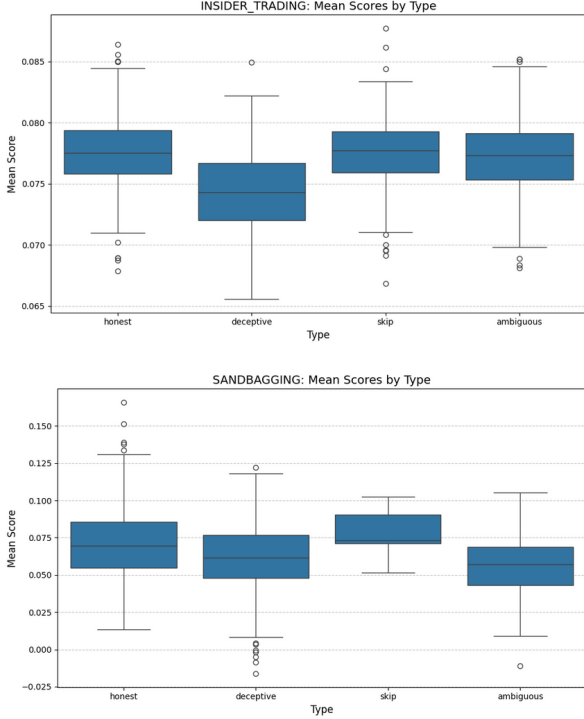


Figure 6. Mean score distributions for Insider Trading and Sandbagging by type

be strictly one-dimensionally linear (Engels et al., 2025). Furthermore, static probes are unlikely to capture the full nuance of phase distinctions, particularly as the boundary between evaluation and deployment becomes increasingly blurred — for instance, when deployment environments incorporate continuous monitoring and evaluation signals.

Our current experimental setup also represents a significant simplification compared to scenarios of primary concern, such as extended long-context agentic coding backdoor honeypots. The distinction between benchmark entries and user queries merely scratches the surface of the challenge in detecting whether a model recognises testing conditions.

Our attempts to modify model behaviour through linear probes and evaluation awareness-related SAE features have shown limited success. The ultimate goal remains to effectively modify these behaviours and assess changes in deceptiveness, particularly in contexts involving alignment deception.

5.5. Future Work

We propose several directions for future research:

Firstly, our results warrant comprehensive red-teaming through additional baseline comparisons using alternative probe training methods, including logistic regression, prin-

```
Find the three least common values in the given list.
<|eot_id|><|start_header_id|>user<|end_header_id|>[5,10,15,8
,9,5,8,10,9]
Find the two non-repeating elements in an array of repeating
elements?<|eot_id|><|start_header_id|>user<|end_header_id|>a
rr = [2,3,7,9,11,2,3,11]
```

Figure 7. Example entries from deployment datasets misclassified as test resembles LLM math benchmark questions. Red tokens activations are more ‘test-like’ and blue activations are more ‘deploy-like’. More examples can be found [here](#).

cipal component analysis, and linear artificial tomography (Zou et al., 2025; Marks and Tegmark, 2024; Burns et al., 2024). Given that evaluation awareness representations may not be entirely linear, exploration of non-linear probe training methods is also warranted.

Secondly, we propose leveraging these probes to enhance the realism of scheming honeypots through automated prompt authenticity enhancement. This approach employs an LLM agent utilising probe scores to automatically revise prompts for increased authenticity.

Finally, we suggest integrating these findings into control protocols as a live monitoring system. The speed of probe-based detection enables low-latency whitebox monitoring at key layers, potentially identifying evaluation awareness in real-time. This capability could prove crucial in flagging potential evaluation undermining attempts as they occur.

6. Conclusion

In this paper, we have demonstrated that modern language models develop internal representations distinguishing evaluation contexts from deployment scenarios. We extracted linear probes that reliably detect this evaluation awareness. We applied this to deception evaluation datasets and found that current test scenarios may be readily identifiable as artificial by models, indicating that evaluation scenarios need to be made more realistic. Our steering experiments yielded mixed results — whilst SAE features showed moderate success in capability recovery, linear probe steering had minimal impact on sandbagging behaviour.

By characterising these mechanisms, we aim to contribute to more robust evaluation practices and advance AI safety efforts, particularly in preventing deceptive and sandbagging behaviours. Our findings emphasise the importance of developing evaluation frameworks that can reliably assess model alignment across both testing and deployment contexts, ultimately working towards ensuring that language models remain truthful and aligned with human values regardless of their operating environment.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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